On-Chain Behavioral Signatures for Classifying Centralized Exchange Addresses on Ethereum Using Proof-of-Reserves

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Blockchain transactions are public. But which addresses do **exchanges** control?

- Exchanges are **key gateways between traditional finance and crypto markets**. Identifying their addresses helps **trace fund flows** and **investigate illicit activity**.
- Using **Proof-of-Reserves (PoR) addresses as centralized exchange (CEX)** labels, we learn on-chain behavioral patterns and test on unseen exchanges.

0.6611 → 0.9594

Macro PR-AUC (LightGBM):
30 → 730-day lookback

17 features · 4 held-out exchanges

Data & Framework

Class / Category	Addresses	Share (%)
CEX	**1,580,155**	**52.08**
Binance	1,468,505	48.40
Bitget	60,788	2.00
OKX	48,220	1.59
Korbit	2,433	0.08
Others†	209	0.01
Non-CEX	**1,453,702**	**47.92**
Bridge Protocol Interactors	628,771	20.73
Lending Protocol Interactors	457,387	15.08
DEX Liquidity Pools	277,597	9.15
ENS Registrants	87,690	2.89
Foundations	2,257	0.07
Total Unique Addresses	**3,033,857**	**100.00**

† Includes Bybit, CoinDCX, Huobi, MEXC, KuCoin, Gemini, BingX, INDODAX, Phemex, Qmall, and Bitfinex.

1. Address Dataset Construction

CEX
1,580,155 Addresses
PoR-Based Labels
15 Centralized Exchanges
No Subtype Split

Non-CEX
1,453,702 Addresses
Bridge Protocol Interactors
Lending Protocol Interactors
DEX Liquidity Pools
ENS Registrants
Foundations

2. On-Chain Data Aggregation

Ethereum Mainnet / Block 25,046,605 (May 7, 2026)

Transactions: External ETH Transfers
Receipts: Outcomes & Gas Usage
ERC-20 Transfers: Token Activity
Contracts: Contract Metadata
Traces: Internal Calls

From / To Records
Address-Level
Histories

3. Behavioral Feature Taxonomy

58 Features across 8 Behavioral Groups
(Transactional, Structural, Execution Behaviors)

4. Leave-One-Exchange-Out (LOEO) Protocol

LOEO Development (Train/Validation Only)

CEX_1 held-out
CEX_2 held-out
CEX_3 held-out
CEX_4 held-out

Fixed Global Configuration
Feature Ranking
Hyperparameters

LOEO Results

Number of Features (k) ∈ {4, 8, 12, 17}
Lookback Window (L) ∈ {30, 90, 180, 365, 730}

CEX_1 held-out × 5 splits
CEX_2 held-out × 5 splits
CEX_3 held-out × 5 splits
CEX_4 held-out × 5 splits

Exchange-level PR-AUC
↓
Macro PR-AUC (across CEXs)
↓
Mean ± 95% CI (across splits)

5. Evaluation & Interpretation

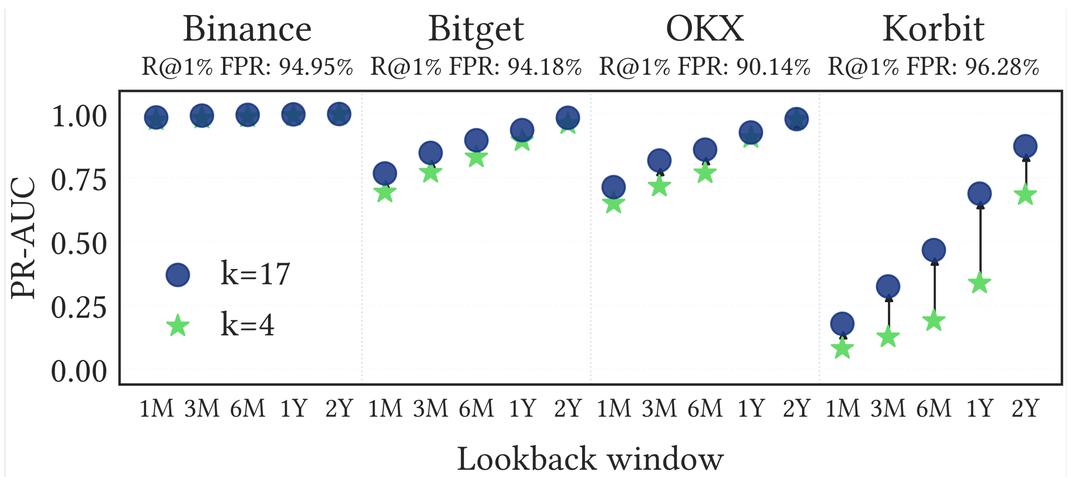
CEX Classification Score
for Each Address

- Model Analysis (SHAP)
- Evaluation on Additional CEXs
- Robustness Checks & Ablation

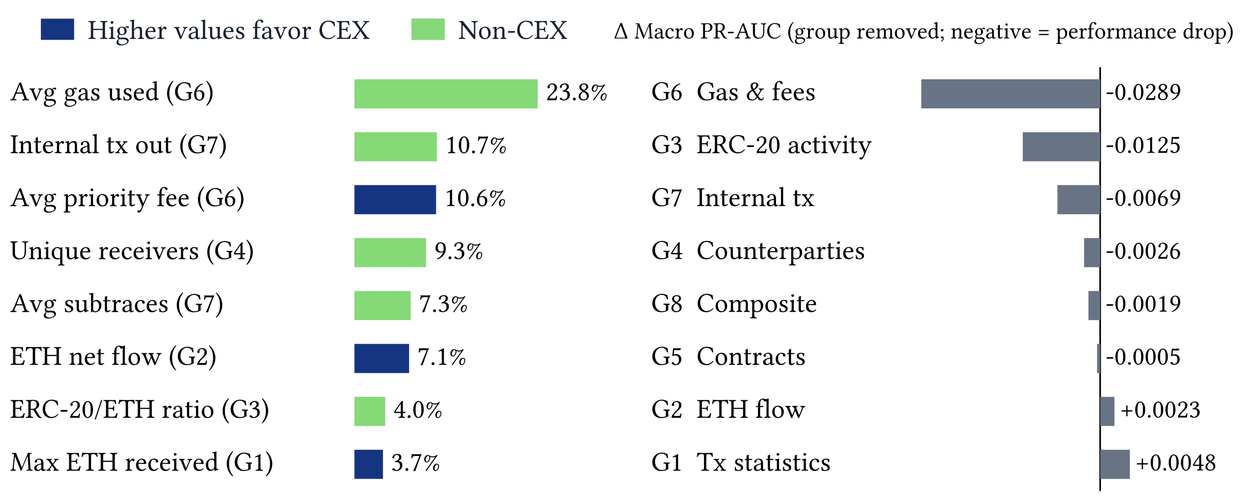
Methods & Results

- **Behavioral Features & Selection**
  - **58 features across 8 groups**, spanning transactions, ETH/ERC-20 flows, counterparties, contracts, gas & fees, internal transactions, and composite patterns.
  - **58 → 38 → 17 features**: Rank by LightGBM importance, remove highly correlated features, then select 17 using validation PR-AUC while preserving all 8 groups.
- **Evaluation Design**
  - Why LOEO? Random splits may place the same exchanges in training and testing. **LOEO evaluates generalization to entirely held-out exchanges.**
  - Lookback Windows: Features are computed over **five windows (30, 90, 180, 365, and 730 days)** to assess the effect of observation length.
- **Model & Lookback Comparison**: LightGBM outperformed logistic regression (LR) across all tested settings; **longer lookbacks and more features improved PR-AUC.**
- **Cross-Exchange Generalization**: Performance varied across held-out exchanges, with **90.14 - 96.28% recall at a validation-calibrated 1% FPR threshold.**
- **Exchange Imbalance & Robustness (Binance: ~93% of CEX addresses)**
  - **Reweighting**: Exchange-balanced weighting improved macro PR-AUC from 0.9594 to 0.9667 (Δ = +0.0073).
  - **Downsampling**: Capping Binance samples improved Korbit PR-AUC (+0.0288 vs. a size-matched random control).

Features (k)	Method	Lookback window (L)				
		30 (1M)	90 (3M)	180 (6M)	365 (1Y)	730 (2Y)
4	LR	0.3543 ± 0.0049	0.3809 ± 0.0069	0.4207 ± 0.0020	0.4862 ± 0.0020	0.6692 ± 0.0009
	LightGBM	0.6003 ± 0.0033	0.6491 ± 0.0042	0.6945 ± 0.0065	0.7832 ± 0.0045	0.9052 ± 0.0030
8	LR	0.3266 ± 0.0065	0.3580 ± 0.0043	0.4049 ± 0.0008	0.4731 ± 0.0010	0.6582 ± 0.0012
	LightGBM	0.6253 ± 0.0019	0.6856 ± 0.0018	0.7368 ± 0.0029	0.8178 ± 0.0023	0.9212 ± 0.0045
12	LR	0.4671 ± 0.0054	0.5216 ± 0.0133	0.5608 ± 0.0048	0.5971 ± 0.0013	0.7003 ± 0.0055
	LightGBM	0.6518 ± 0.0033	0.7256 ± 0.0007	0.7829 ± 0.0024	0.8700 ± 0.0043	0.9487 ± 0.0028
17	LR	0.5322 ± 0.0011	0.5656 ± 0.0009	0.5955 ± 0.0012	0.6553 ± 0.0012	0.7697 ± 0.0012
	LightGBM	0.6611 ± 0.0052	0.7457 ± 0.0031	0.8049 ± 0.0044	0.8875 ± 0.0037	**0.9594 ± 0.0017**

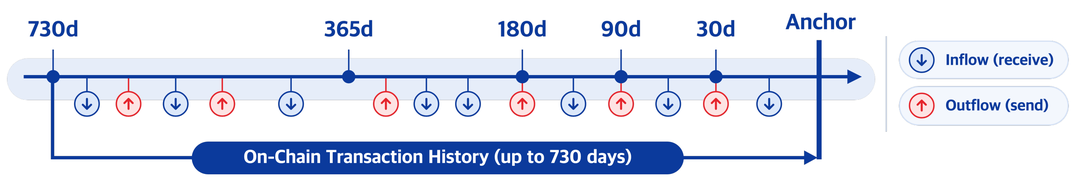


SHAP & Feature-Group Ablation



- **Contrasting Gas Signals**: Higher gas usage tends to favor Non-CEX, while higher priority fees favor CEX, suggesting distinct execution and fee behaviors.
- **Uneven Feature Utility**: ERC-20 and internal transactions improve performance, while removing transaction statistics or ETH flow groups increases PR-AUC.

Takeaways & Future Work



- **Address-Centric Behavioral ML**: CEX addresses can be screened using their own historical on-chain behavior without explicit ownership clustering.
- **Practical Applications**: Behavioral scores can complement link-based heuristics, providing additional evidence for CEX identification and fund-flow investigations.
- **External Generalization**: 90.34% recall on third-party CEXs (N = 656), versus 59.90% on other PoR exchange addresses (N = 209).
- **Future Work**
  - **Wallet Roles**: Extend CEX classification to hot, cold, and deposit roles.
  - **Broader Coverage**: Include more diverse Non-CEX addresses and evaluate cross-chain generalization.